

ADRIANA HOLTZMAN

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OBJECTIVE

Engineer passionate about research that helps people – particularly in embedded systems, computer architecture, edge AI, IC design, and graphics. Eager to contribute to applications in biotechnology, hardware design, robotics, and art.

EDUCATION

Carnegie Mellon University , Pittsburgh, PA Master of Science in Electrical and Computer Engineering	December 2027
Carnegie Mellon University , Pittsburgh, PA Bachelor of Science in Electrical and Computer Engineering	May 2027 GPA: 3.87/4.00
École Polytechnique Fédérale de Lausanne , Lausanne, Switzerland Exchange student with Computer Science concentration	Spring 2026

RESEARCH & WORK EXPERIENCE

Embedded Systems Laboratory, ERIP Researcher , Lausanne, Switzerland · Selected for Excellence in Research Internship Program at EPFL for summer 2026 advised by Dr. David Atienza Alonso · Work on LZ4 compression techniques for energy efficient die-to-die interconnects between edge AI chips using SystemVerilog	March 2026 - August 2026
Draper Labs, Undergraduate Systems Engineer , Cambridge, MA · Wrote unit tests for Inertial Measurement Unit data analysis application to support acceptance testing in MATLAB · Verified tests in Linux environment and deployed code to GitLab CI/CD Pipeline to be utilized during all merge requests · Devised scalable approach to track utilization across 15+ Acutronic Rate Tables using C and ACUTROL Control Language	May 2025 - August 2025
Optimal Route Generation Pipeline for CircumNav Rover , Pittsburgh, PA · Developed novel methodology for future CircumNav Rover NASA proposal advised by Dr. William (Red) Whittaker · Used BFS, DFS, A*, and DBSCAN Clustering in Python to find and optimize circumlunar routes across binarized terrains · Identified and collaboratively solved coordinate system distortion bug to ensure viability of outputted routes	January 2025 - August 2025
Micro Robotics Lab, Actuator Design for Fish-like Robots , Pittsburgh, PA · Lead project to create caudal fin actuators for autonomous fish-like robots, advised by Dr. Sarah Bergbreiter · Employed Solidworks, Onshape, Arduino, rapid prototyping, and electrical design to develop inertia-based turning capability · Designed safe, reusable test setup to isolate electronics above water tank and track fin location underwater	May 2024 - August 2024

TEACHING EXPERIENCE

Embedded Systems (18-349), Teaching Assistant , Pittsburgh, PA · Lead small groups, graded assignments, and held office hours for collaborative project-based course · Coached students in building PCB, designing Real Time OS kernel in C, and conceptualizing embedded systems content	August 2026 - Present
TechSpark Machine Shop, Teaching Assistant and Technician , Pittsburgh, PA · Guided shop users to operate machines safely and assisted professional shop machinists in manufacturing parts · Instructed 50+ students and tested curriculum by machining C-clamp as Teaching Assistant for manual machining courses	May 2024 - Present

SKILLS

Languages	C/C++, Python, SystemVerilog, x86-64/ARM Assembly, MATLAB, LaTeX, HTML/CSS
Software	Git, Cadence Virtuoso, VCS, Vivado, VSCode, Autodesk Fusion, KiCad, SolidWorks, Onshape, Arduino
Machines	Mill, Lathe, CNC Mill, Waterjet, Laser Cutter, 3D Printer (FDM/SLA), Bandsaw
Spoken	English (Native speaker), Spanish (CEFR B1), French (CEFR A2)
Hobbies	Music (Choir, Guitar, Listening, Synthesizers), Jewelry Making, Machining, Woodworking

RELEVANT COURSEWORK

Circuits Digital IC Design Electronic Devices & Analog Circuits Signals & Systems	Architectures Structure & Design of Digital Systems Embedded Systems Computer Systems	Robotics Advanced Computer Graphics Space Robotics Development Fundamentals of Control
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PROJECT EXPERIENCE

Extension to NORI Physics-based Renderer

February 2026 - May 2026

- Built final project in C++ for CS-440 Advanced Computer Graphics at EPFL extending NORI renderer to render cave scene
- Contributed heterogeneous participating media, texture/normal mapping, depth of field, and environment map lighting

Real Time Operating System Embedded Kernel

August 2025 - December 2025

- Built kernel on STM32 Cortex M4 using custom PCB designed in Autodesk Fusion to operate miniature car
- Implemented context switching, mutexes, RMS scheduling, interrupt-based UART, and I2C in C and ARM Assembly
- Designed PID controller in user program to drive wheel motors at constant speed regardless of disturbances

GPS Real Time Kinematics (RTK) Data Collection and Analysis

January 2024 - December 2025

- Lead data collection and analysis for annual University Raceday 2025 (carbon fiber gravity racing) as CIA Data Chair, utilizing high-precision GNSS data to advise driver lines, push team selection, and energy loss analysis
- Trained team of 10+ people to collect and process RTK position data and contribute to data processing tools

Malloc Memory Allocator

July 2025

- Optimized implementation in C to include segregated free list, footerless allocated blocks, and 16 byte miniblocks
- Achieved 73.8% Utilization and 5605 Kops/sec Throughput across 20 traces, significantly reducing internal fragmentation

FPGA Serial Communication Hardware Thread

February 2025

- Built FSM and datapath in SystemVerilog, VCS, and Vivado to receive, decode, and transit serial hamming codes
- Accounted for inputted bit rates within 5% of nominal 12580 Hz by using edge detection to enable resynchronization

Analog Audio Synthesizer, Build18 Hardware Hackathon

October 2024 - January 2025

- Manufactured analog synth with filter, echo, and oscillator functionalities for compatibility with instruments and mic inputs
- Presented product and live demonstration to team of corporate sponsors as 1 of 5 selected Innov18 teams

EXTRACURRICULAR EXPERIENCE

Carnegie Involvement Association (CIA Buggy), RD25 Data Chair, Mechanic, Build

August 2023 - Present

Build18 Hardware Hackathon, Web Development Officer

February 2025 - May 2026

EPFL Watchmakers, Machinist

February 2026 - May 2026

Scotch'n'Soda, Actor, Co-Head Paint Implementer, Asst. Electrician

August 2023 - May 2024

Counterpoint A Cappella, Vocalist

August 2023 - May 2024